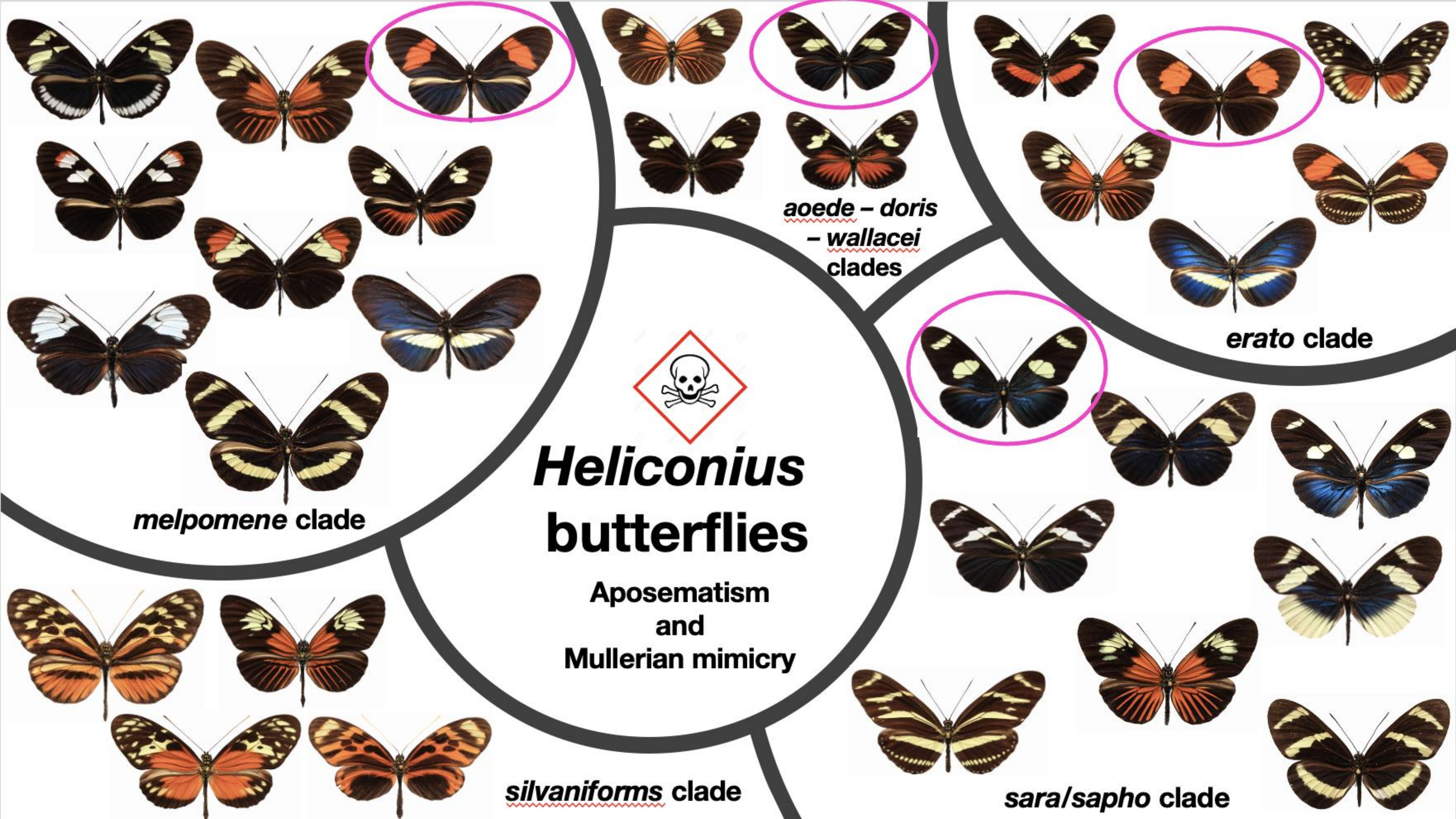
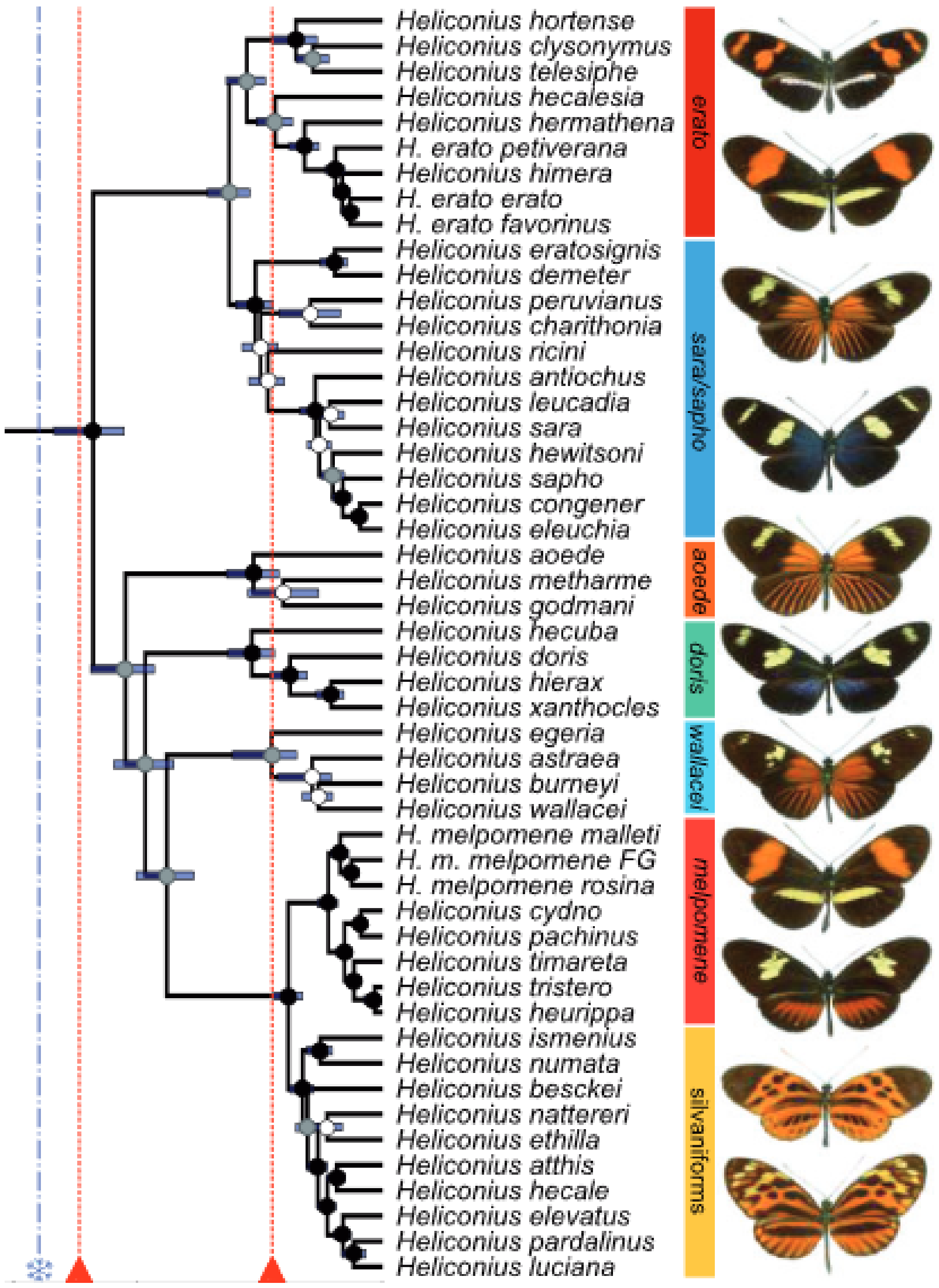
# Introduction to *Heliconius* butterflies

## Biodiversity genomics course

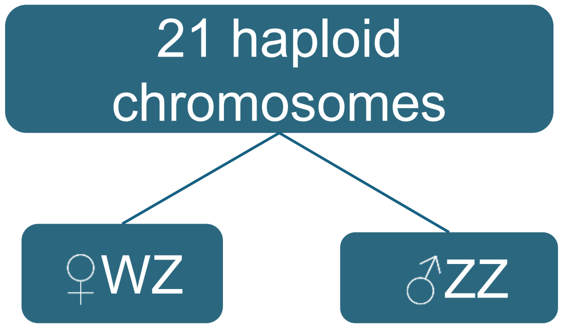
Bogotá-Colombia
July. 2026


aoede – doris – wallacei clades
erato clade
melpomene clade
Heliconius butterflies
Aposematism and Mullerian mimicry
silvaniforms clade
sara/sapho clade



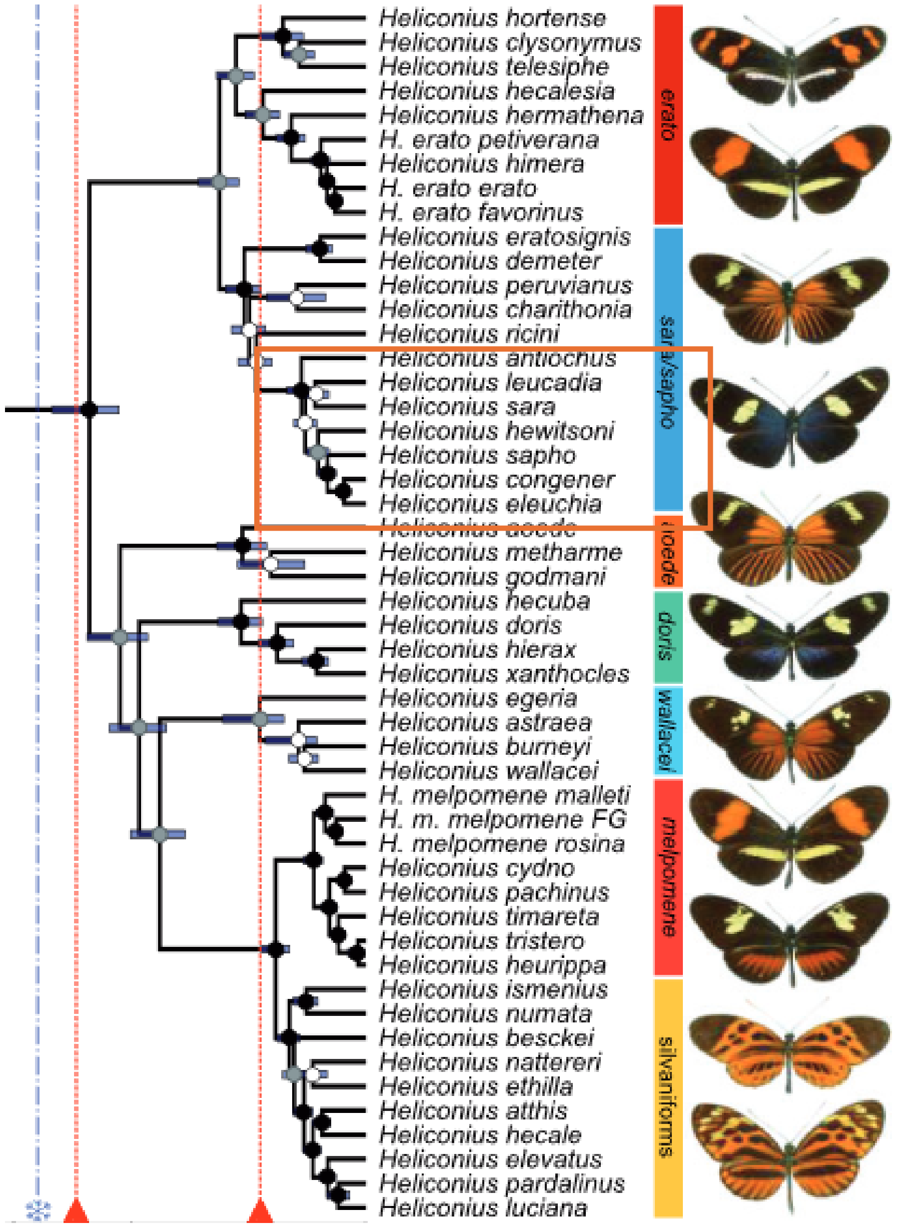

Kozak, 2015

# Neotropical *Heliconius* butterflies




**Sex chromosome system**

# Introduction



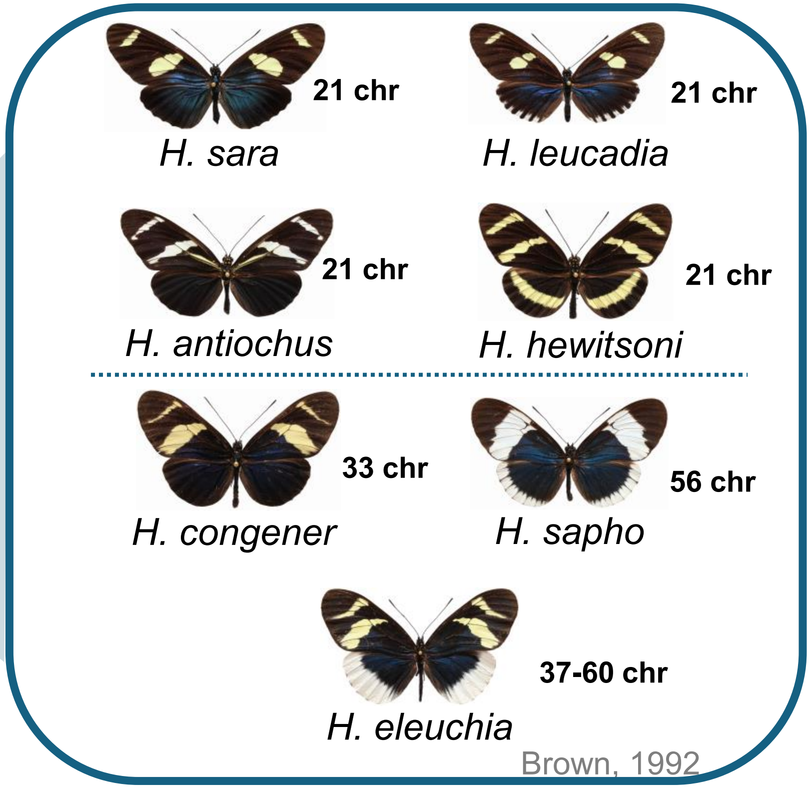

Kozak, 2015

21 - 60 chromosomes

Sex chromosome system

♀$W_1W_2W_3Z_1Z_2Z_3$

♂$ZZ_1ZZ_2ZZ_3$



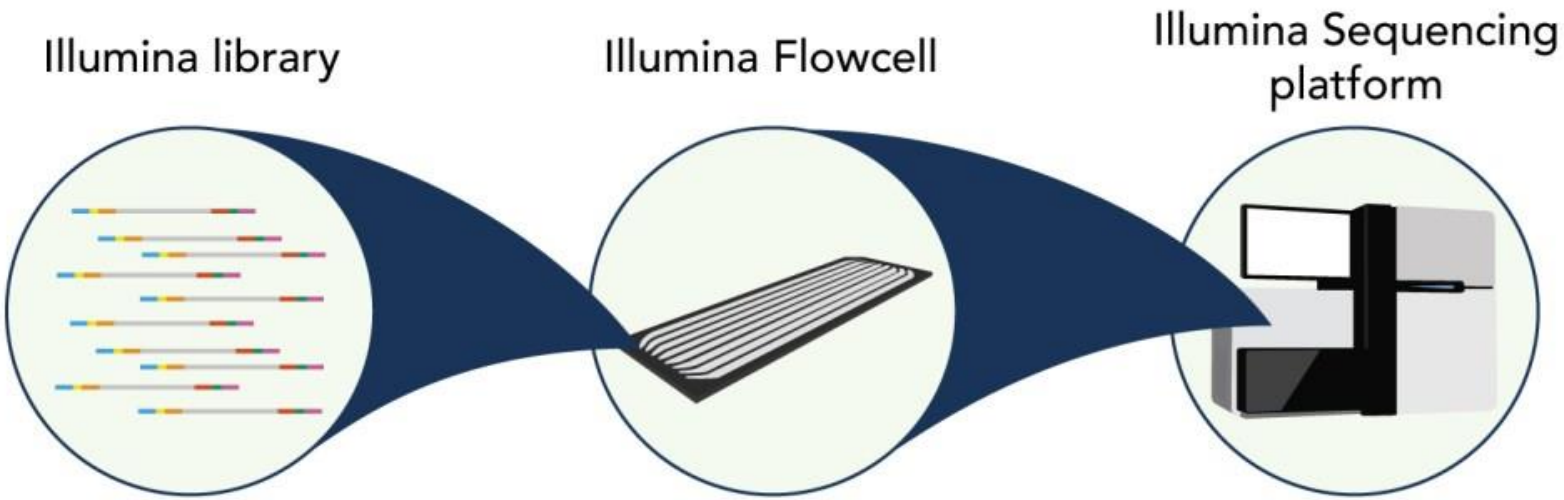

The data (**raw reads**) we will be working with today and tomorrow are from this clade.

How do raw sequences files look like?

How can we do a quality control of them?

# Raw sequences are store in FASTQ files

1. Extracting high-quality DNA.
2. Repairing the DNA ends (if needed).
3. Attaching sequencing adapters (and barcodes if multiplexing).
4. Loading the prepared library onto the flow cell for sequencing.



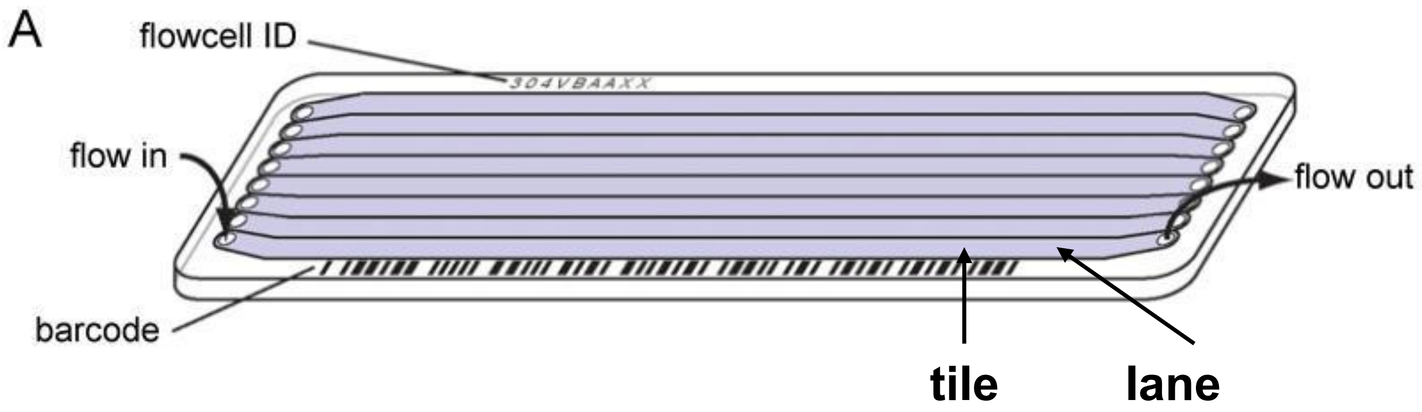

https://joshuapenalba.com/illumina-sequencing/

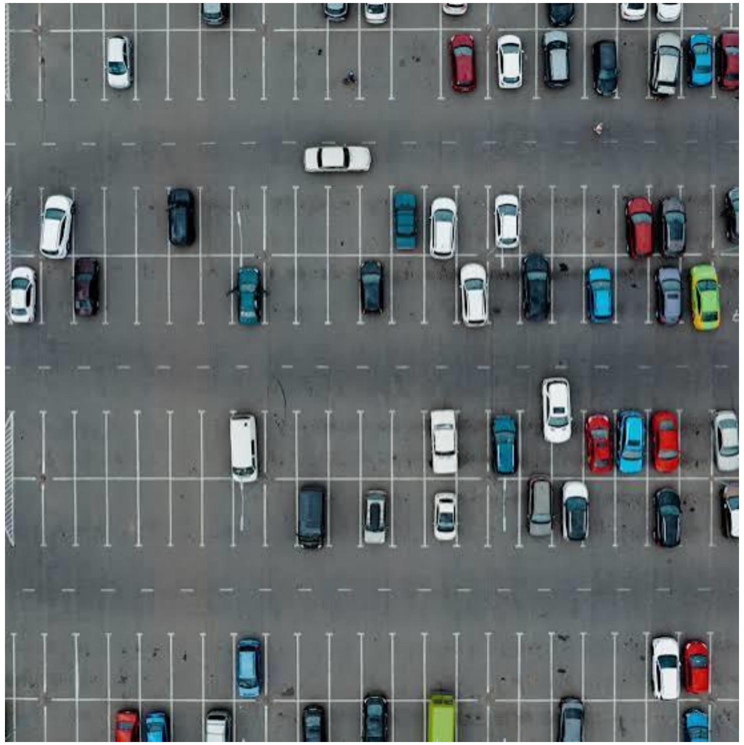
A FASTQ file is the standard way to store millions of short DNA fragments, this format include:

- the DNA sequence *and*
- how confident the machine was about each base it read.

# What is a flow cell?

- **Lane** = a large section of the flow cell where sequencing takes place.
- **Tile** = a small region within a lane that the sequencer images separately.



- The **flow cell** is the entire parking lot.
- Each **lane** is a row of parking spaces.
- Each **tile** is a small group of parking spaces within that row.
- Each **DNA cluster** is a car parked in a space.

# FASTQ files extension could be *.fastq* or *.fq*. If compressed should be *.fastq.gz* or *.fq.gz*

## Each of the 4 lines will represent a read.

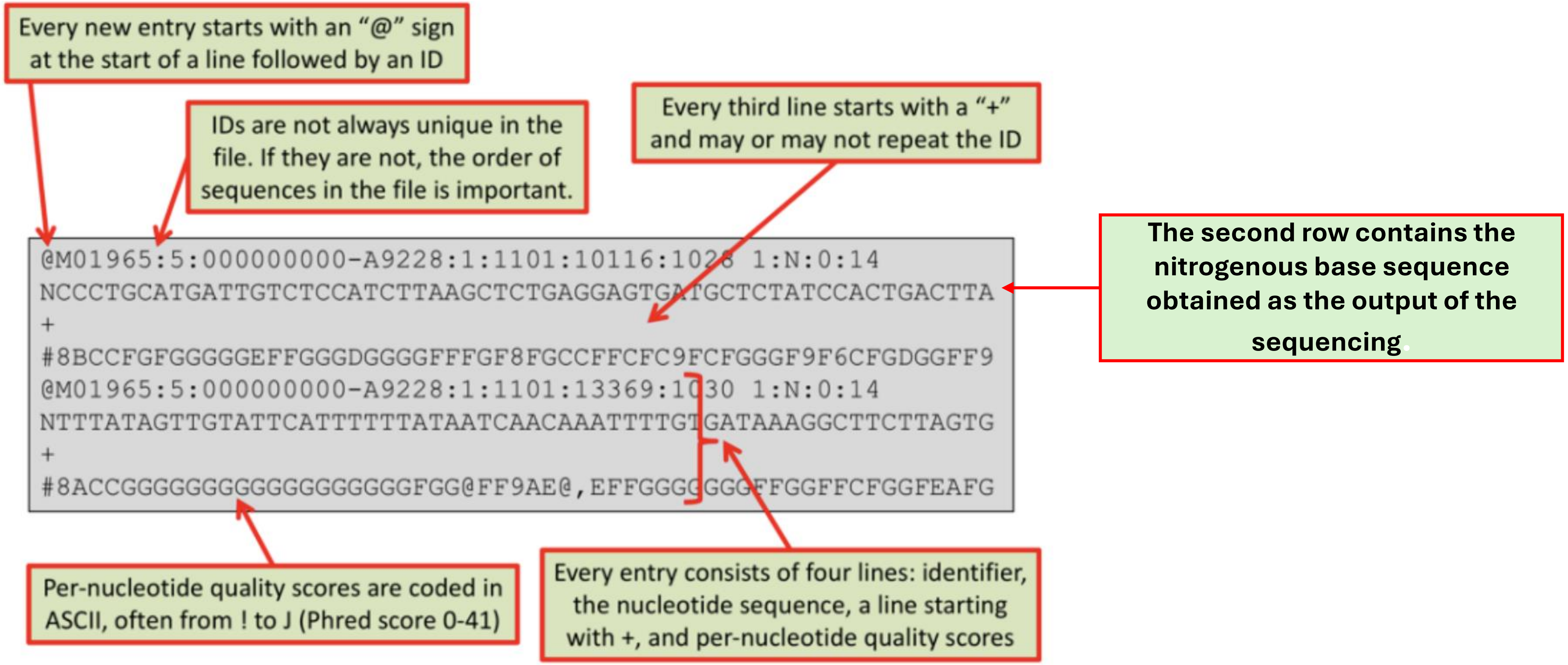



Figure 1. Structure of a FASTQ file. Source: "Theoretical Biology and Bioinformatics" course, Utrecht University, Bas E. Dutilh & Can Keşmir

# **Read header** starts with an @ and includes **sequencing metadata**

## Illumina read header format

colon

@A00469:53:H7VFYDSX2:1:1101:1122:1000 1:N:0:AGTCAA

| Field | Example | Description |
|---|---|---|
| Instrument name | A00469 | Sequencer ID |
| Run number | 53 | Run identifier |
| Flowcell ID | H7VFYDSX2 | Flowcell barcode |
| Lane | 1 | Lane number on flowcell |
| Tile | 1101 | Tile within the flowcell |
| X-coordinate | 1122 | Cluster X-position |
| Y-coordinate | 1000 | Cluster Y-position |
| Read number | 1 | Read pair (1 = R1, 2 = R2) |
| Is filtered | N | Y/N for passing Illumina chastity filter |
| Control number | 0 | Internal control indicator |
| Index sequence | AGTCAA | Sample index/barcode |

**In paired-end sequencing:**

- Read 1 and Read 2 will have headers that **only differ in the read number** (1:N: vs 2:N:).
- Tools like fastp, bowtie2, or bwa use this info to recognize and synchronize pairs.

# FASTQ files quality scores


Figure 2. Detailed structure of a FASTQ file.
Source: https://www.drive5.com/usearch/manual/fastq_files.html

**Table 1. ASCII characters from ! to J (ASCII values 33-74) encode Phred quality scores from 0 to 41. This corresponds to a probability of $10^{-0}$ (100%) to $10^{-4.1}$ (<0.01%) that the nucleotide at that position is wrong. Phred+33 encoding is e.g. used by traditional Sanger sequencing and by the latest Illumina machines.**

| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ! | 33 | 0 | ' | 39 | 6 | - | 45 | 12 | 3 | 51 | 18 | 9 | 57 | 24 | ? | 63 | 30 | E | 69 | 36 |
| " | 34 | 1 | ( | 40 | 7 | . | 46 | 13 | 4 | 52 | 19 | : | 58 | 25 | @ | 64 | 31 | F | 70 | 37 |
| # | 35 | 2 | ) | 41 | 8 | / | 47 | 14 | 5 | 53 | 20 | ; | 59 | 26 | A | 65 | 32 | G | 71 | 38 |
| $ | 36 | 3 | * | 42 | 9 | 0 | 48 | 15 | 6 | 54 | 21 | < | 60 | 27 | B | 66 | 33 | H | 72 | 39 |
| % | 37 | 4 | + | 43 | 10 | 1 | 49 | 16 | 7 | 55 | 22 | = | 61 | 28 | C | 67 | 34 | I | 73 | 40 |
| & | 38 | 5 | , | 44 | 11 | 2 | 50 | 17 | 8 | 56 | 23 | > | 62 | 29 | D | 68 | 35 | J | 74 | 41 |

| ASCII character | ASCII value | Phred score |
|---|---|---|

Figure 3. ASCII characters. Source: "Theoretical Biology and Bioinformatics" course, Utrecht University of
Bas E. Dutilh & Can Keşmir

Usually Phred scores are between **0**, which means an error rate of **100%**, and **41**, which is a probability of **0.01%** that the nucleotide is wrong, and therefore an error.

Transforming the Phred score into ASCII allows us to have only one digit per quality value. This reduces the size of the file and allows a 1:1 match between nucleotide and quality score.

# How can we assess quality of the Fastq files?

## Why assess quality?

Because raw sequencing data often contains:

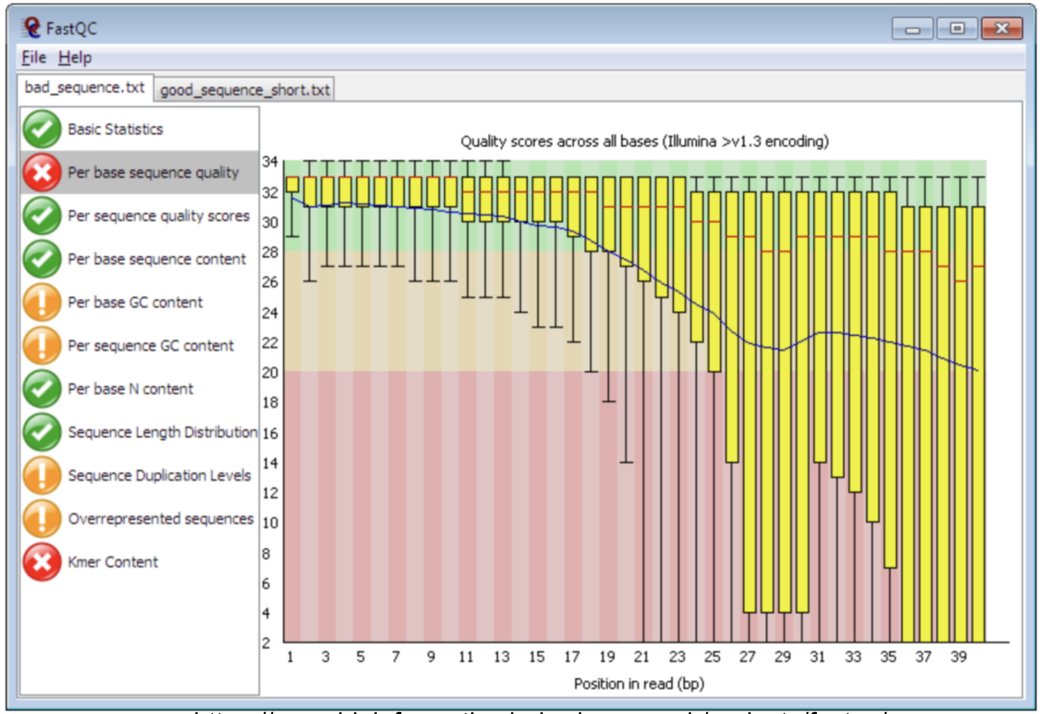
- ➢ Low-quality bases, especially at read ends
- ➢ Adapter contamination
- ➢ Overrepresented sequences (PCR artifacts)
- ➢ Biased GC content
- ➢ Unusual base composition
- ➢ Incomplete or failed reads

## Poor-quality reads can:

•Bias alignment

•Create false variants

## FastQC software



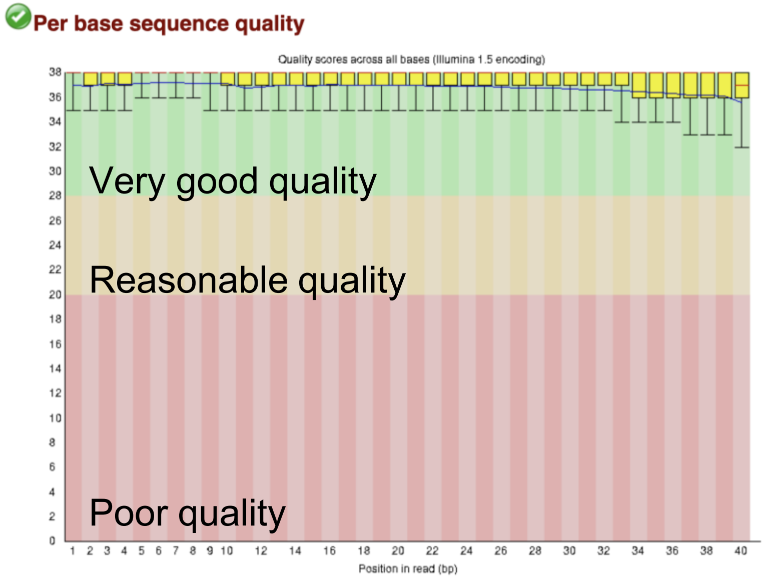

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

```
fastqc -t 4 sample1.fastq.gz
                              OR

fastqc -t 4 -o ./fastqc_results/ /path/to/fastqs/*.fastq.gz

cd fastqc_results/
multiqc .
```

Let’s have a look at the first few sequences and check the sequencing quality with fastqc

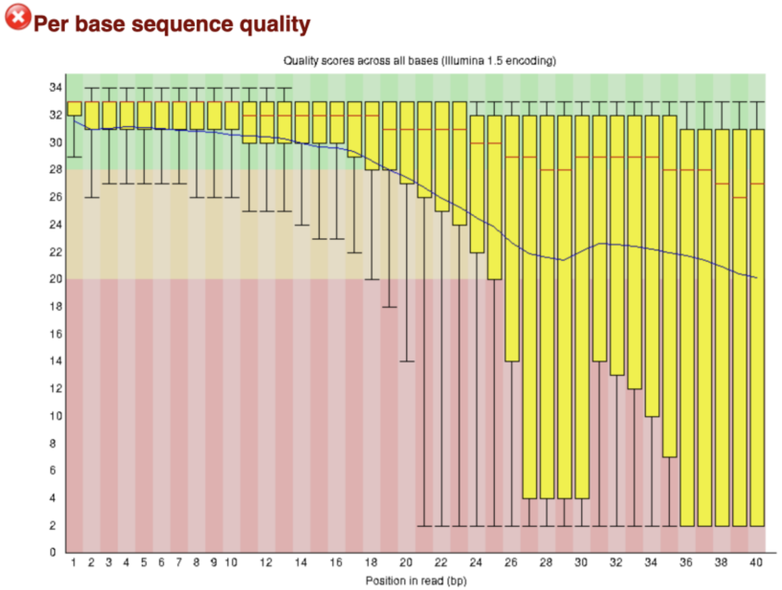
# Interpreting FastQC results: Sequence quality by position

### Good sequence




- The higher the score, the better the base call.
- The quality of reads on most platforms will drop at the end of the read. This is often due to signal decay or phasing during the sequencing run.

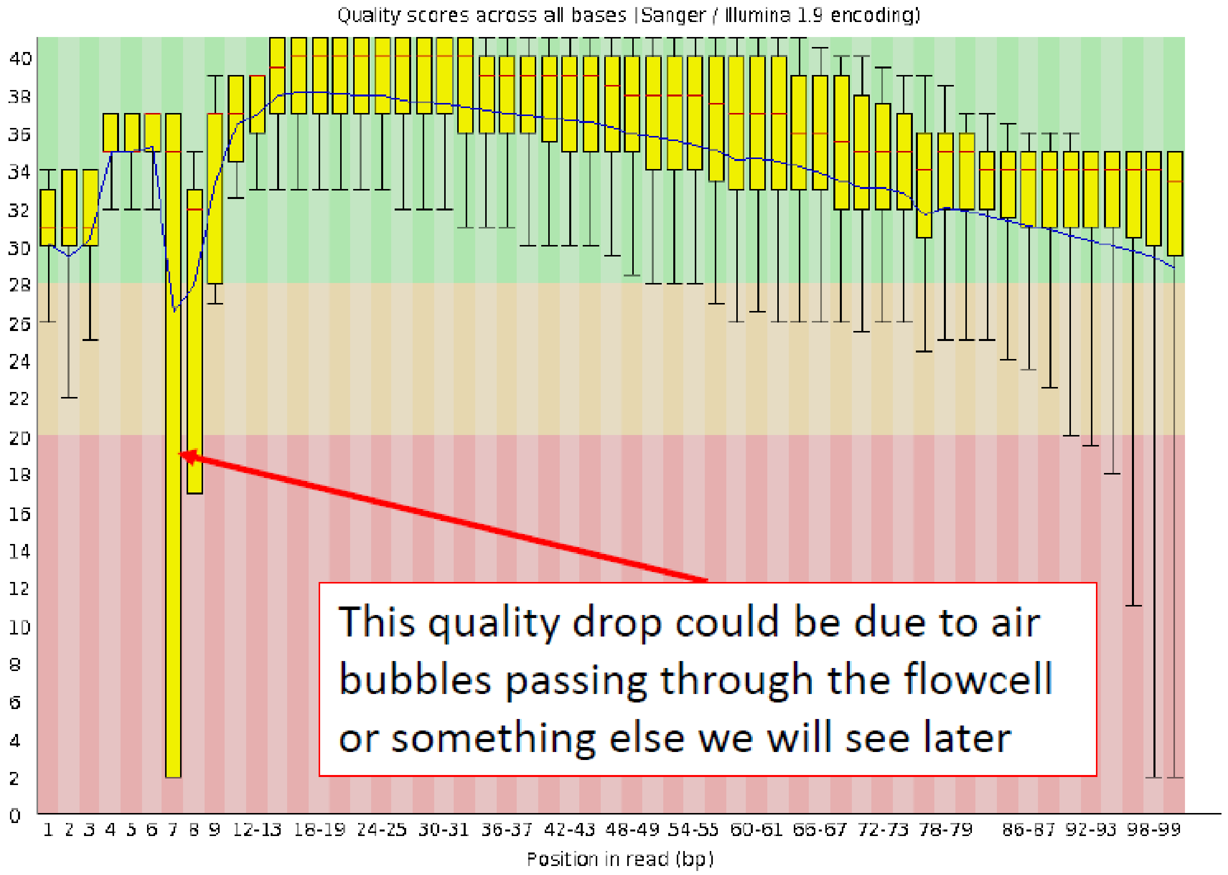
### Bad sequence



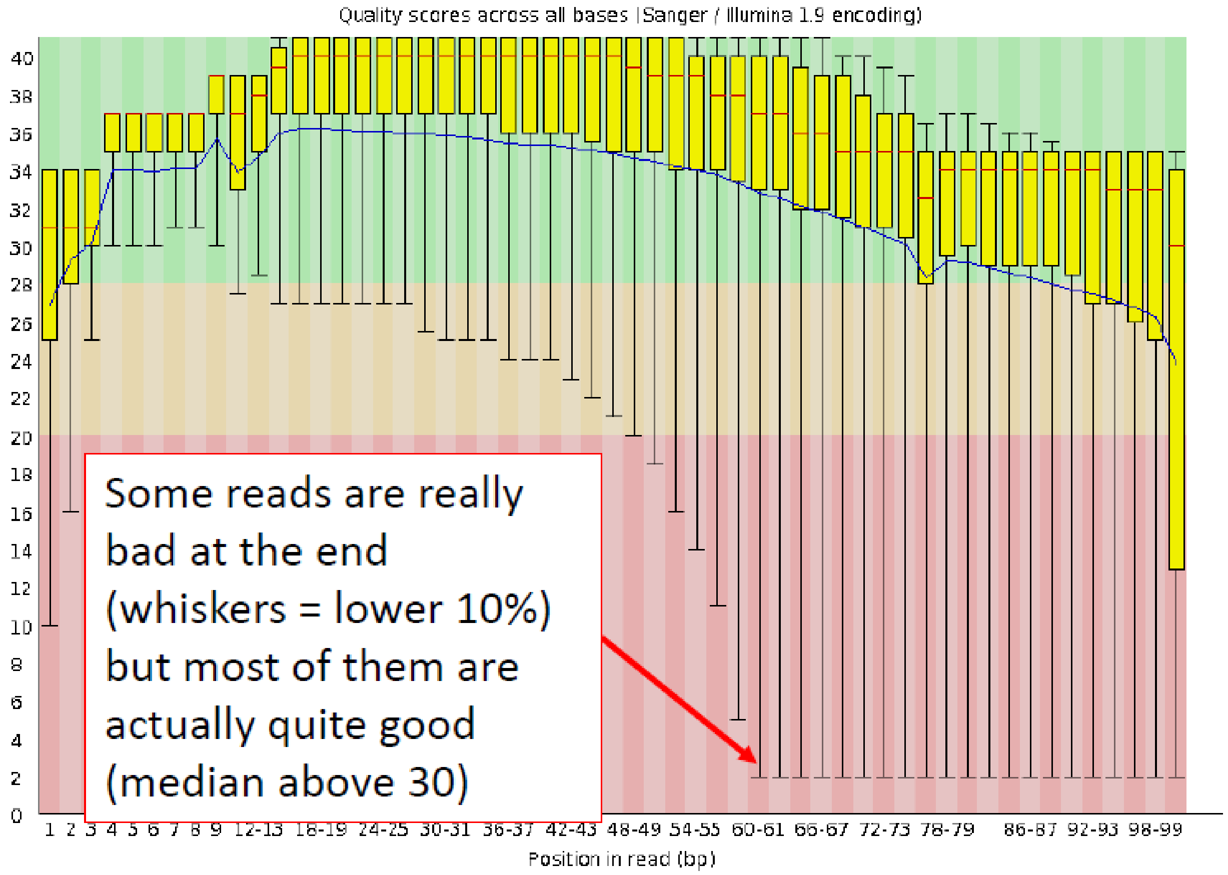

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

# Quality scores across bases: RAD datasets

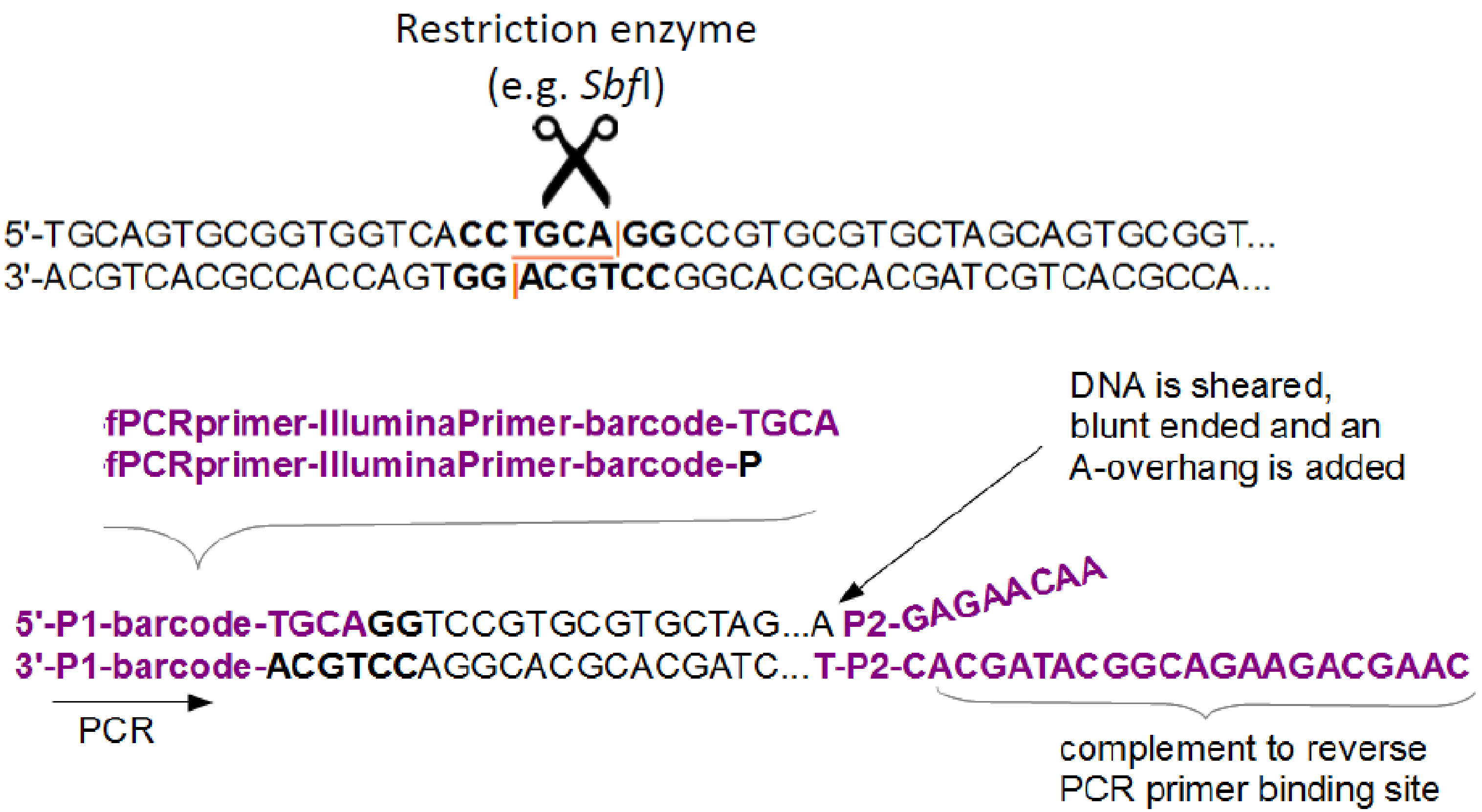
**RAD1**




**RAD2**



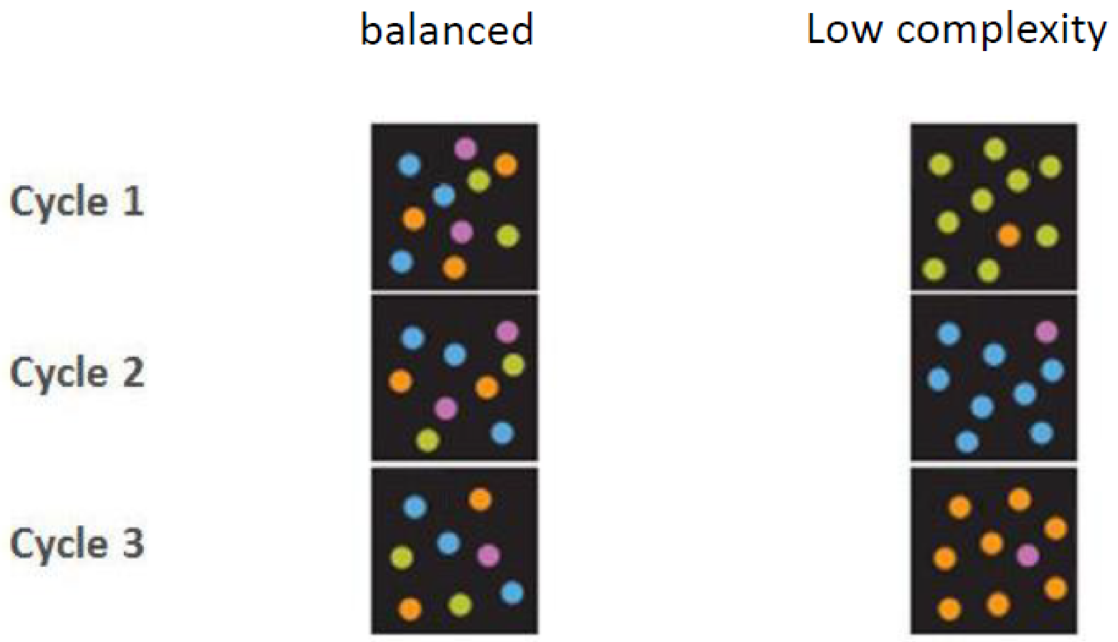
# RAD/GBS




**Each read: will start with the barcode, then the restriction site, then a variable sequence**

# Issues with cluster identification

Due to low complexity at the beginning of the sequence,
Illumina cannot distinguish if a signal comes from one or two clusters

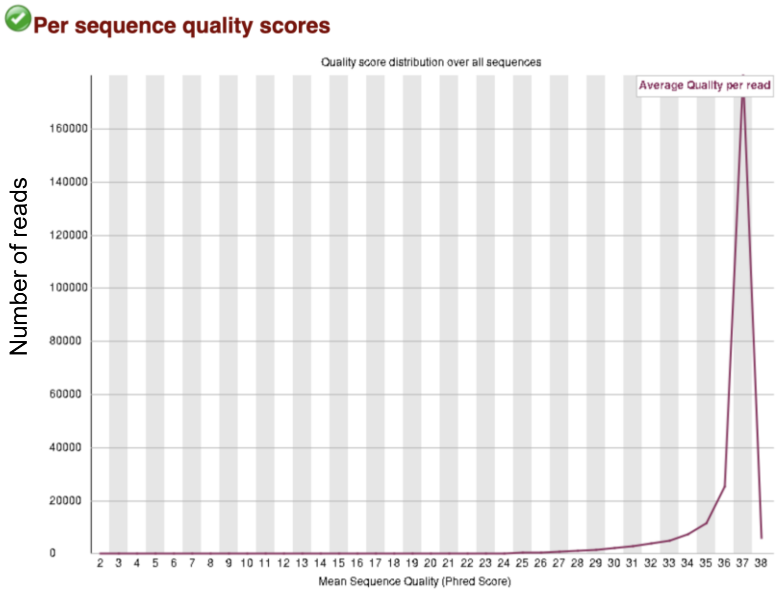


# How to minimize the problem

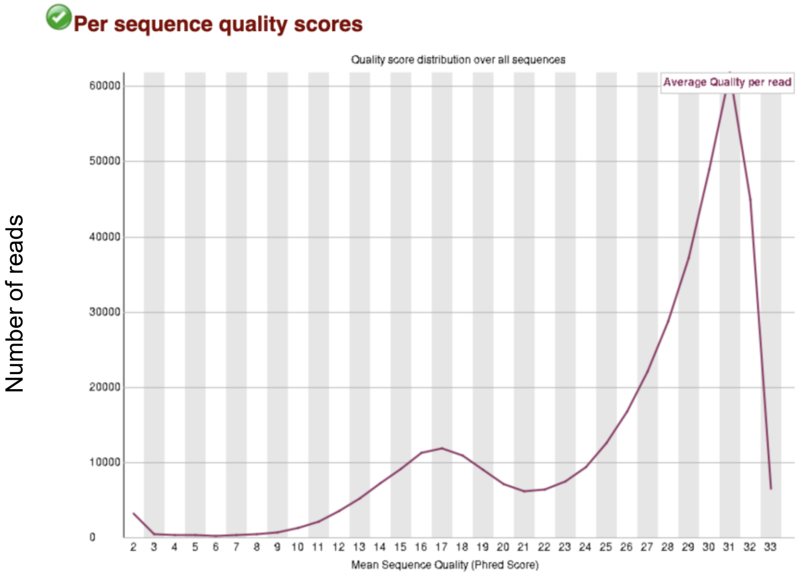
- Use barcodes of different lengths to shift the restriction enzyme cut site
- Add PhiX virus DNA to the RAD libraries to increase the complexity of reads (‘spiking’)
- Reduce loading concentrations of Illumina plates
- Potentially: filter out bad reads

# Interpreting FastQC results: Per sequence quality score and per base sequence content

## Good sequence

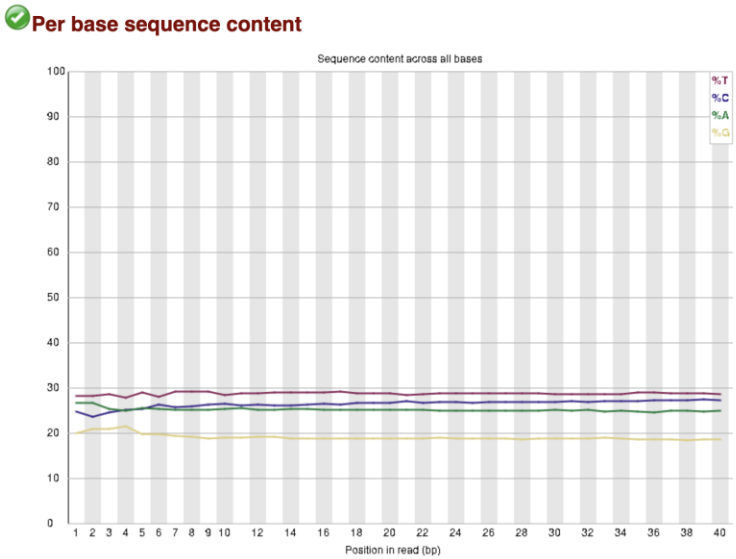



## Bad sequence




### Per sequence quality

- Indicates **heterogeneous quality**: some reads are very high quality, but a sizable fraction is lower.
- **Trimming** or filtering reads with low mean quality.
- Almost all reads are very high quality.
- Tight distribution centered around Phred 36–38.



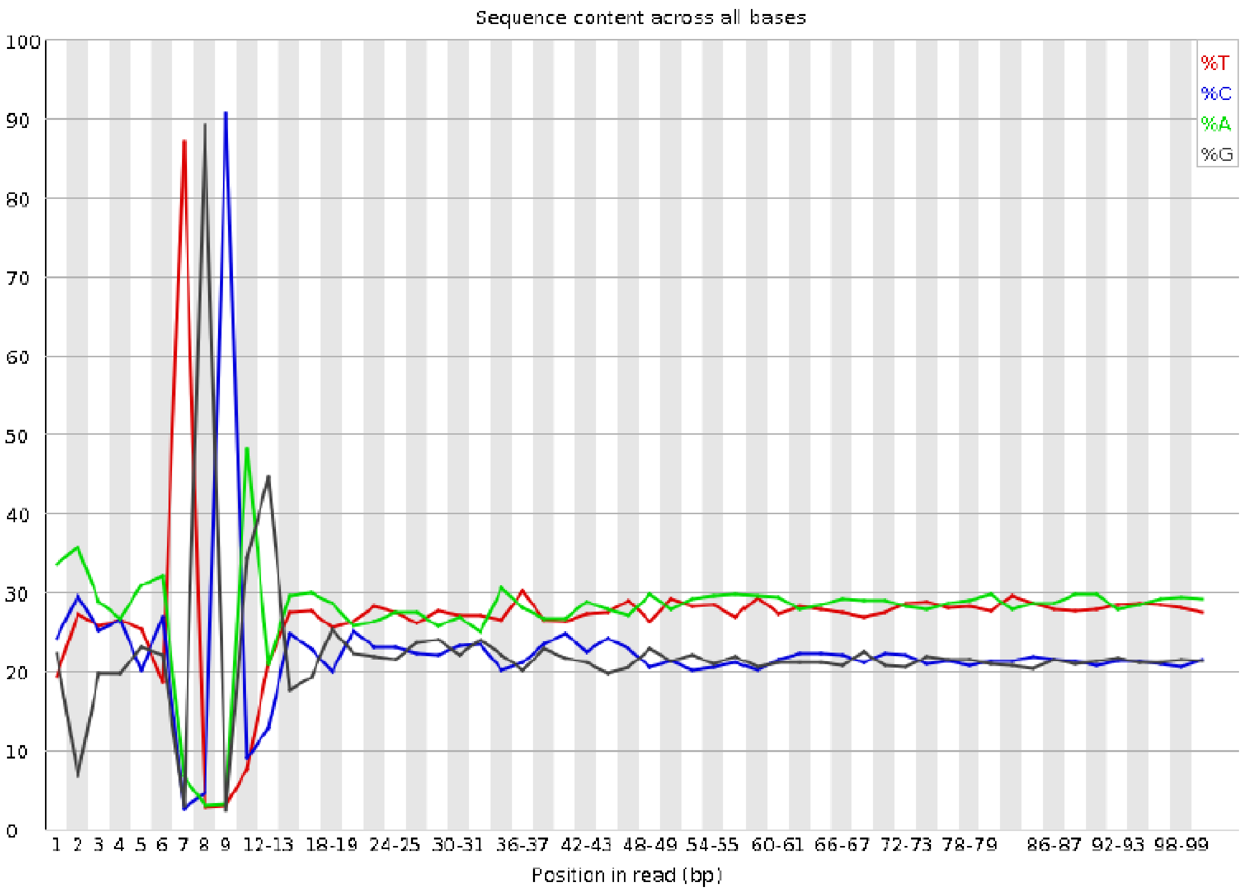

### Per base sequence content

- Proportion of each nucleotide (A, T, C, G) at each position in the reads.
- These plots help detect **sequence composition bias**, which can arise from library preparation artifacts, adapter contamination, or non-random fragmentation.

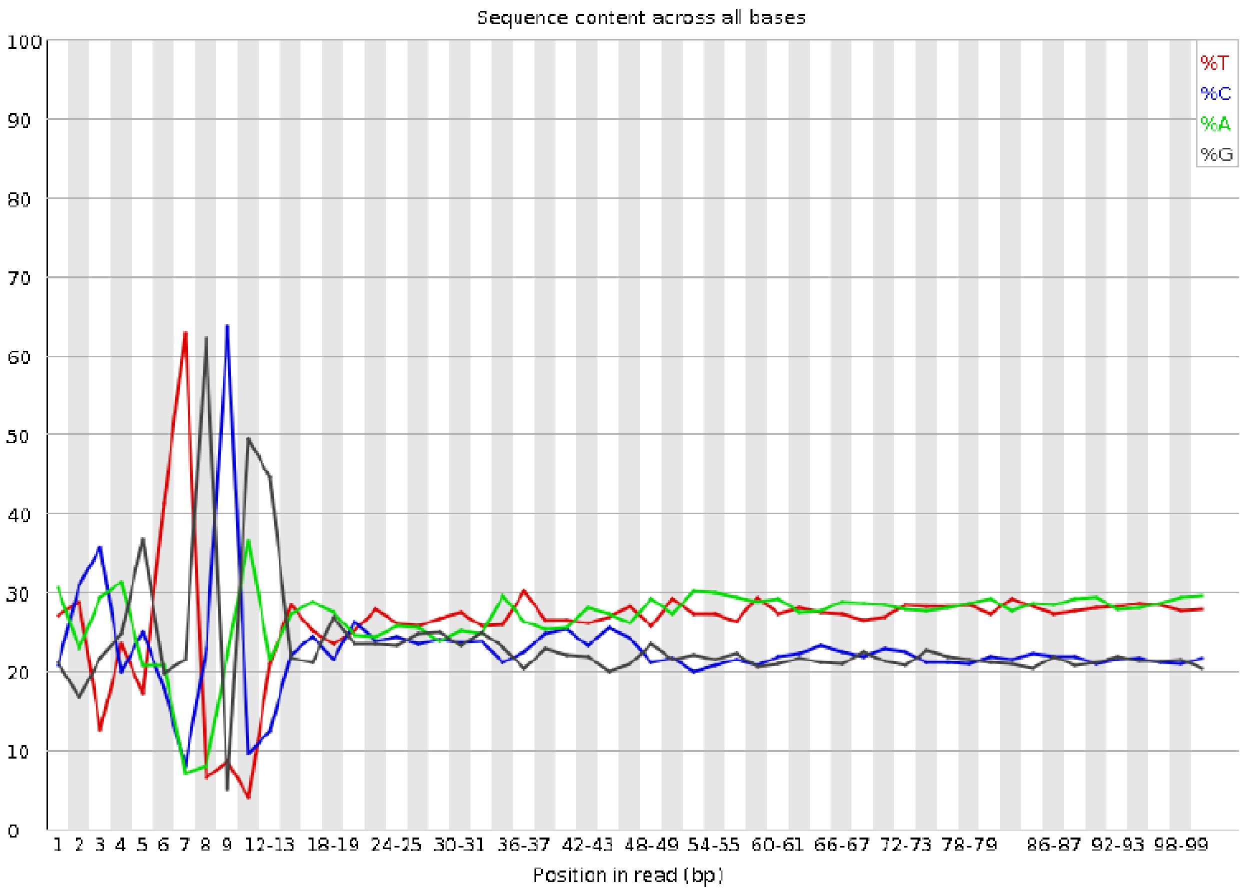
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

# Per base sequence content

**RAD1**

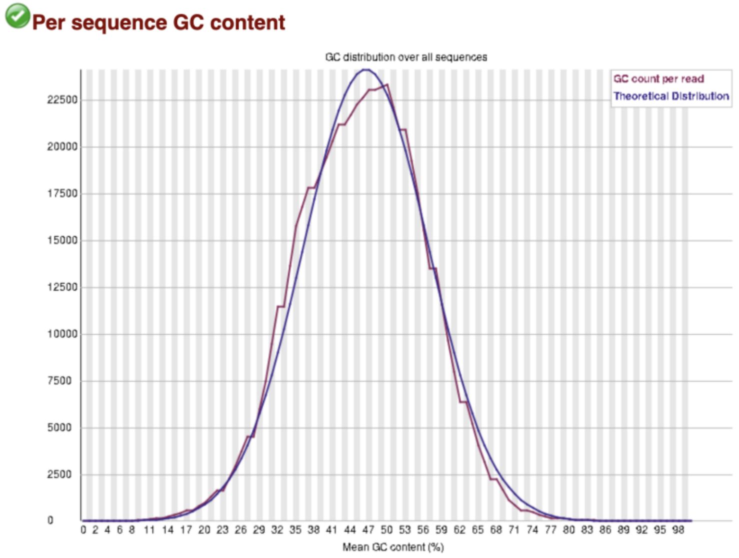



**RAD2**



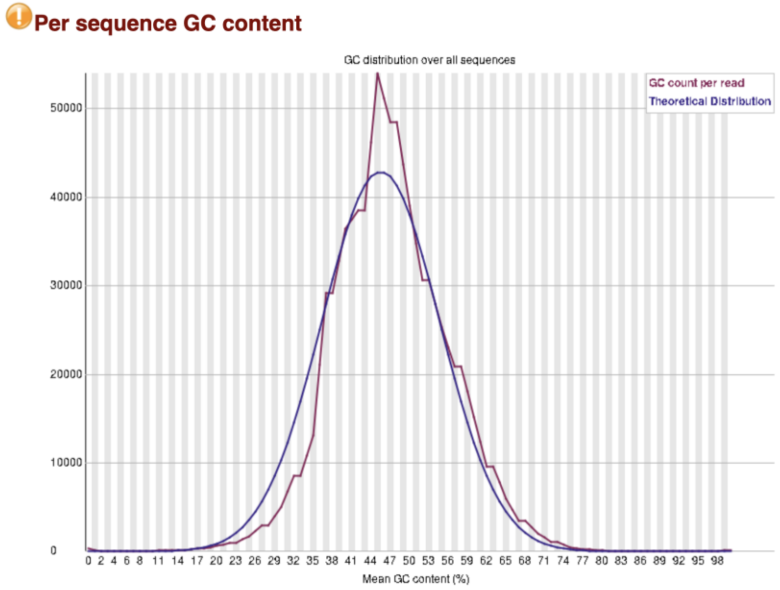
# Interpreting FastQC results: per sequence GC content

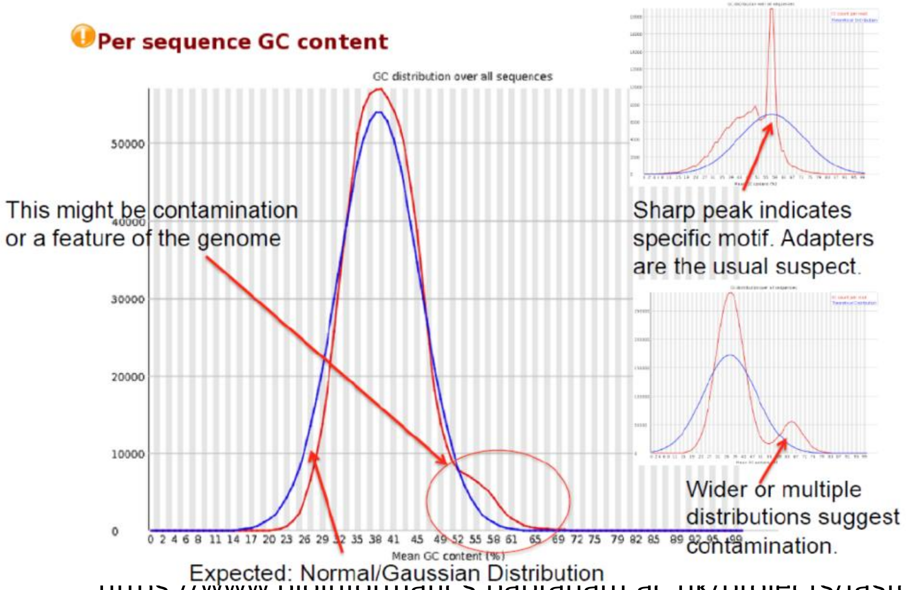
## Good sequence




➢ These figures show a comparison betwee the **observed GC content distribution (red)** of your sequencing reads to an **expected (theoretical) distribution** (blue).

## Bad sequence



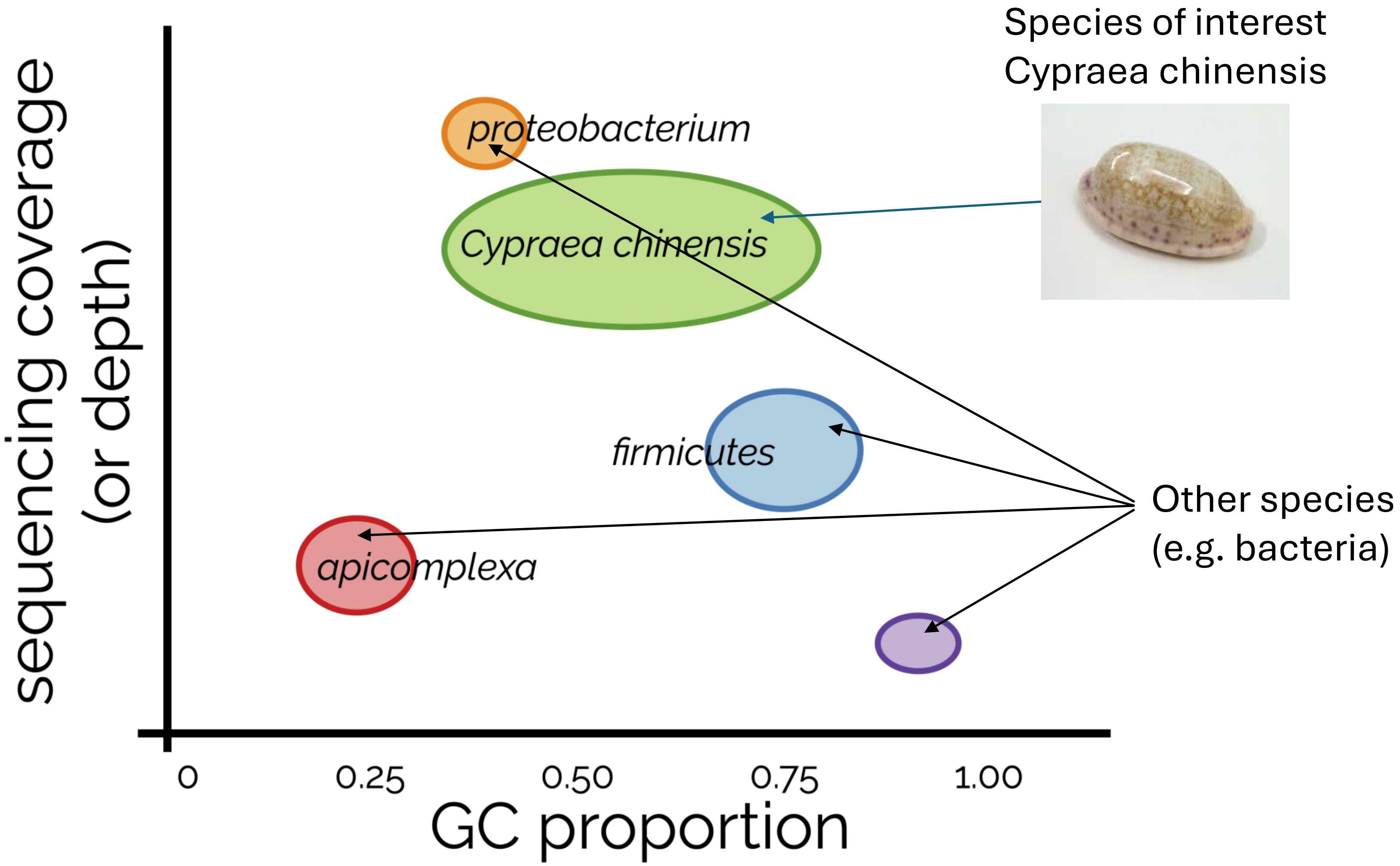




- **Red line** = observed distribution in your actual data.
- **Blue line** = theoretical/expected distribution (a normal/Gaussian curve, calculated from the average GC content of your data).

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

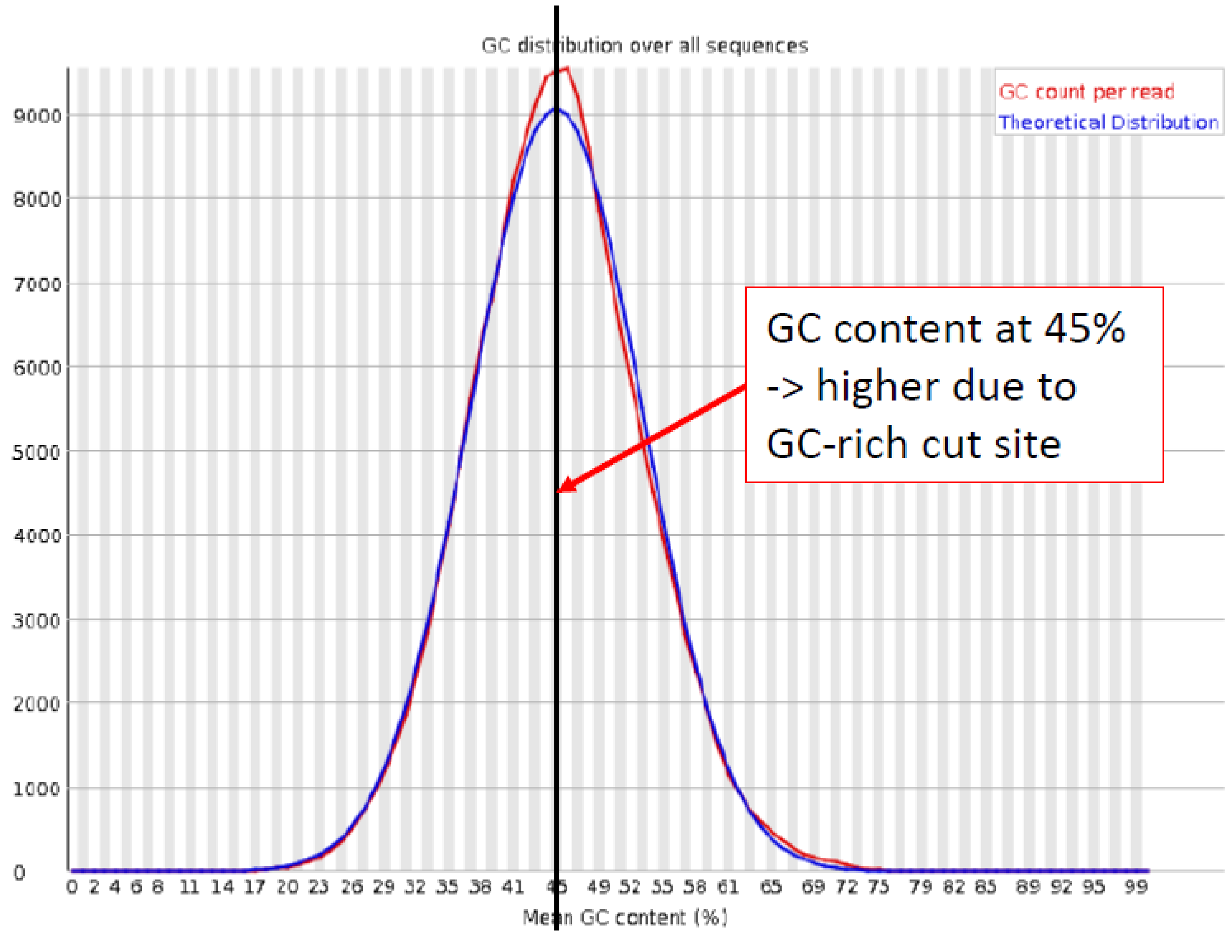
# Blobtools kit uses the GC versus AT proportion to identify contamination from bacteria or other organisms

(https://blobtoolkit.genomehubs.org/)

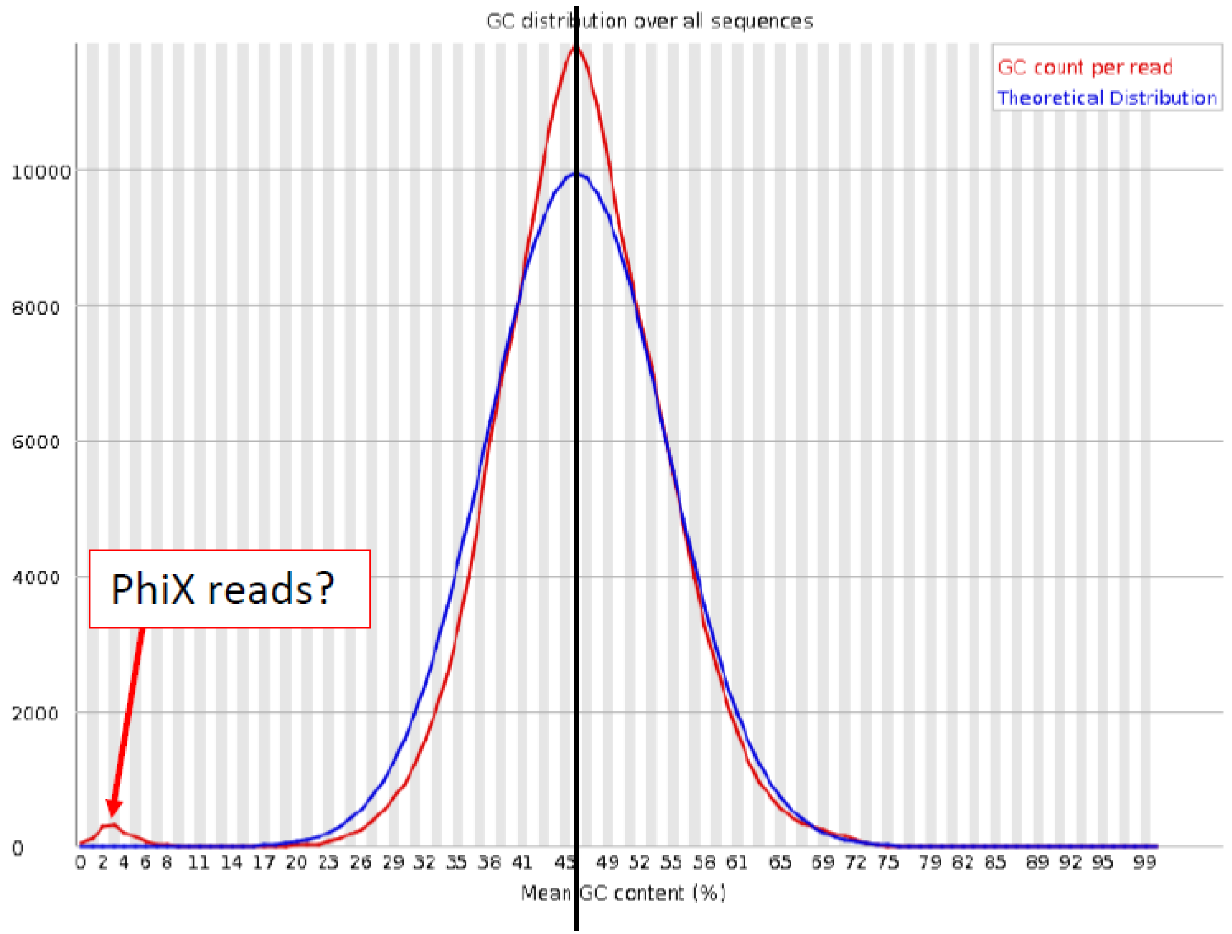


# GC distribution over all sequences

**RAD1**



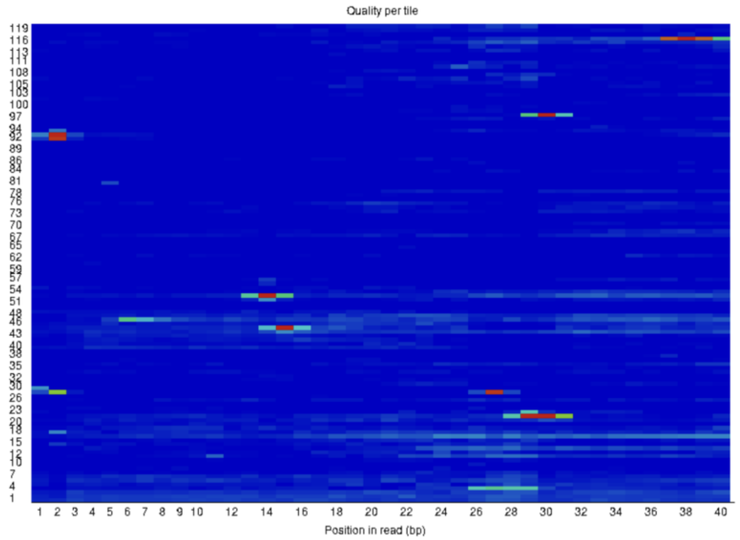

**RAD2**



# Interpreting FastQC results: Per tile FastQC and Per base N content

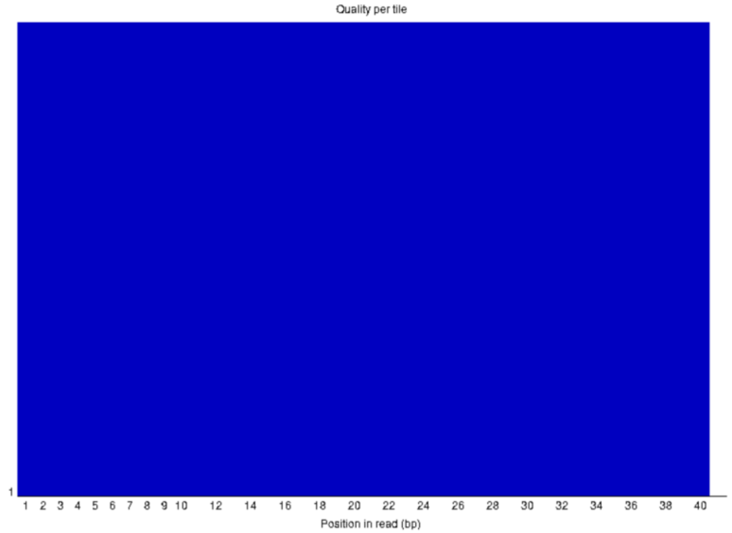
## Slightly less good sequence

**Per tile sequence quality**



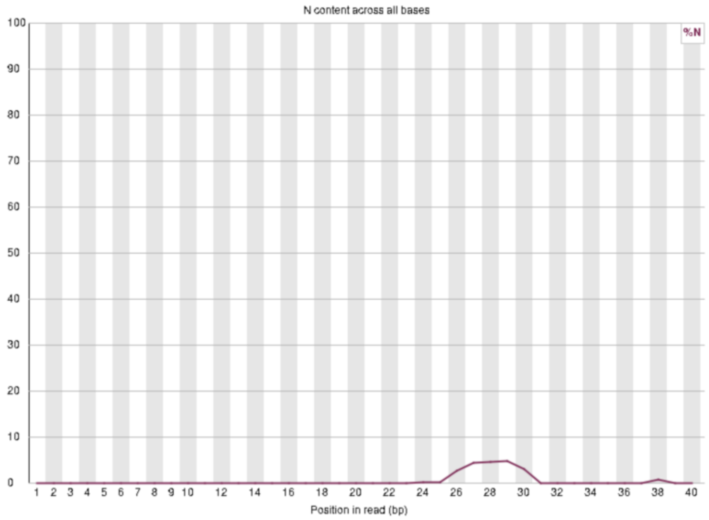

## Good sequence

**Per tile sequence quality**



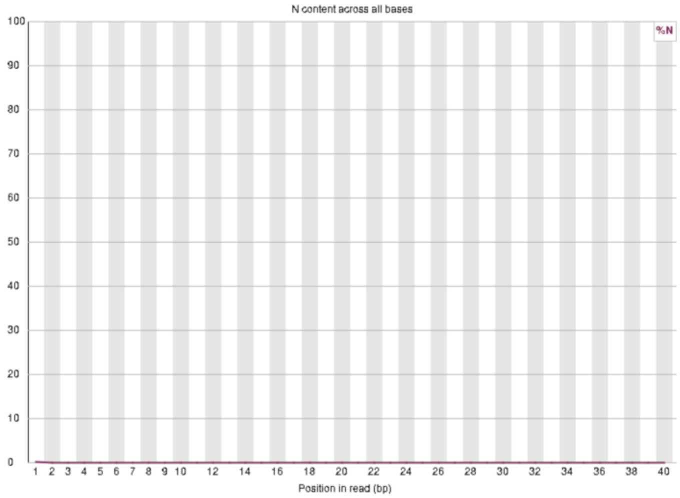

- ➢ Quality scores from each tile across all of your bases to see if there was a loss in quality associated with only one part of the flowcell.
- ➢ The hotter colours indicate that reads in the given tile have worse qualities for that position than reads in other tiles.

**Per base N content**



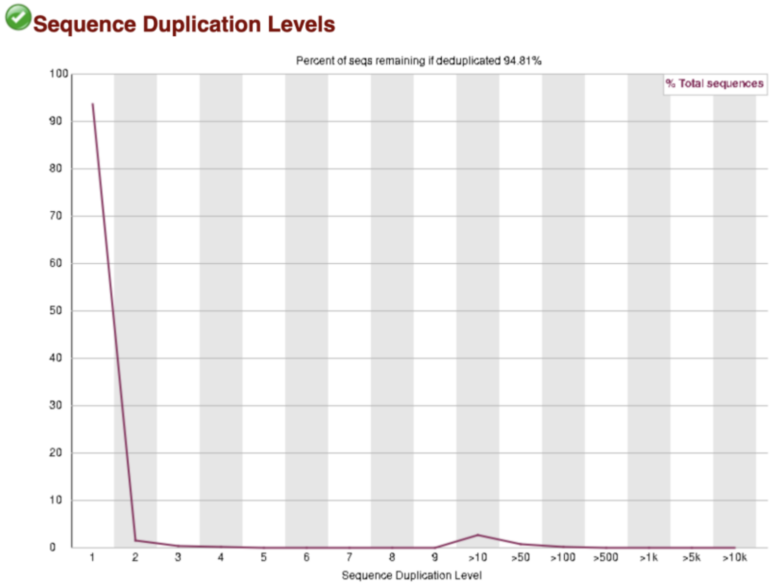

- ➢ This figure reports the **percentage of ambiguous bases ("N")** at each position along the reads.
- ➢ This can result from:
  - Low signal intensity
  - Mixed signals
  - Sequencing chemistry failure
  - Base-calling uncertainty

**Per base N content**

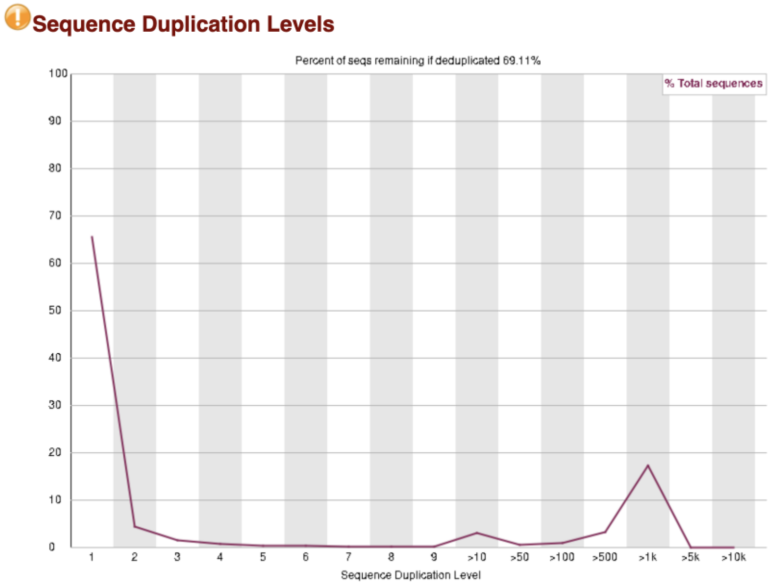


# Interpreting FastQC results: Sequence duplication levels and sequence length distribution

## Good sequence



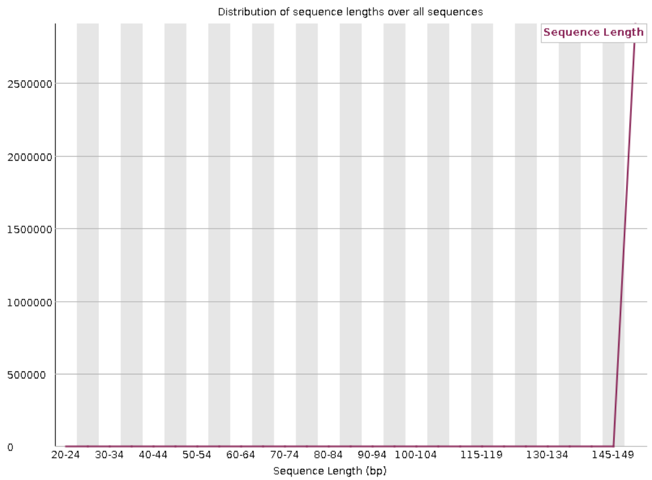

## Bad sequence




- ➢ This figure shows how many **duplicate sequences** exist in your data.

- ➢ A lot of duplication could mean:
- Overamplification during PCR
- Low input sample
- Technical artifact, such as adapter dimers or contamination

## Sequence length distribution



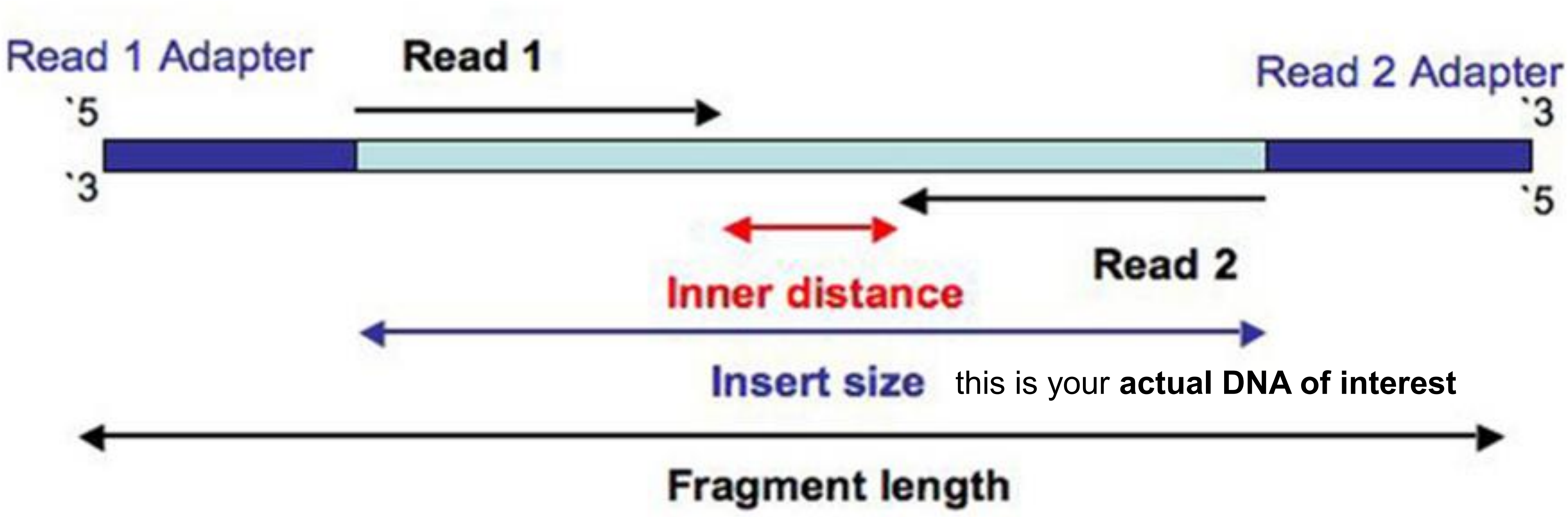

**Sequence length distribution (should really say fragment length distribution)**

- ➢ This plot shows the distribution of fragment sizes in the file which was analysed.

- ➢ Ideally, it will show a single peak at the highest length (150 bp), which means that the forward and reverse reads do not overlap and therefore it could not calculate the fragment length

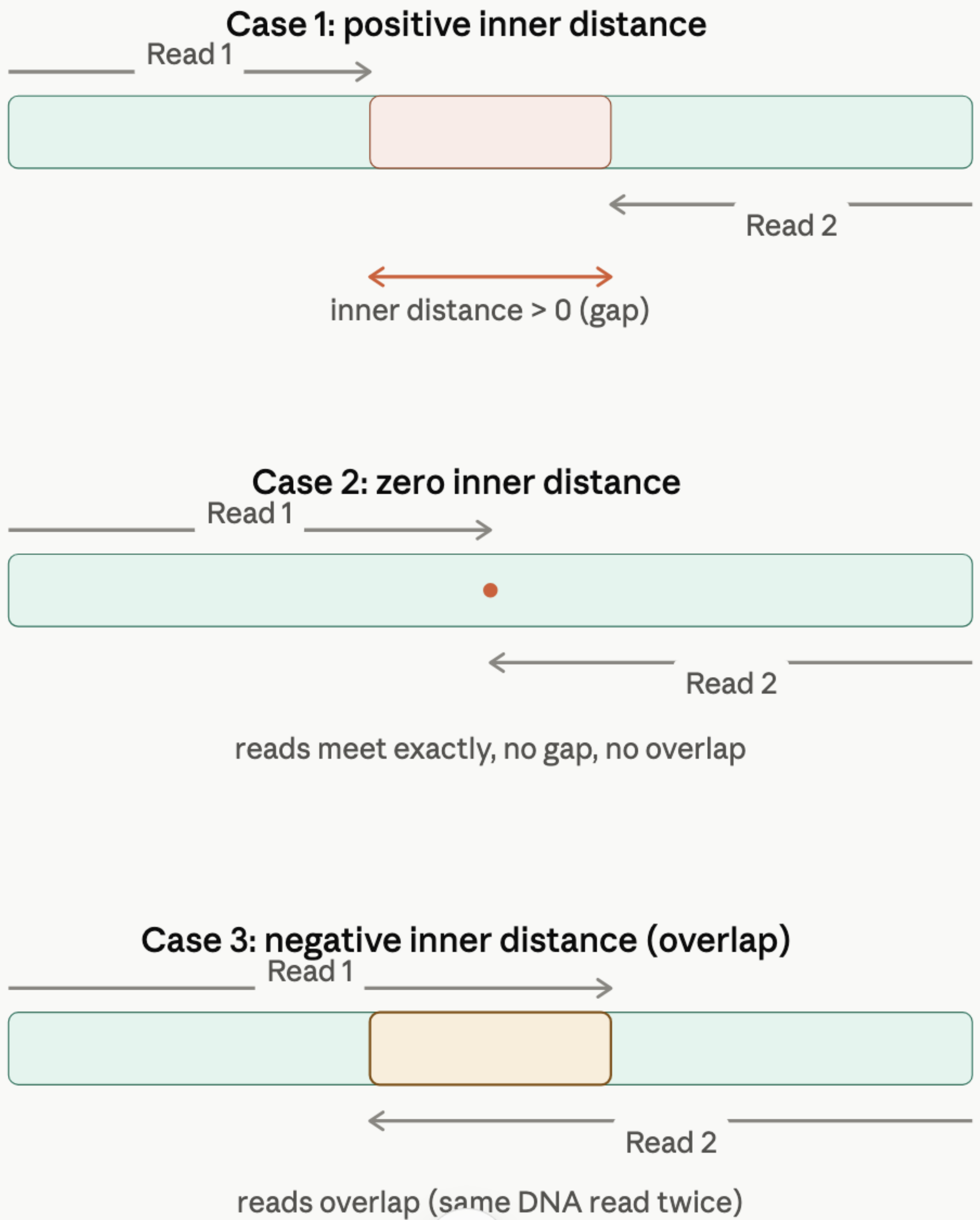
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

# Ideally the two reads should not overlap like here




this is your **actual DNA of interest**

The total length of the DNA molecule that gets sequenced, including both adapters.




**If the reads overlap, fastqc can calculate the fragment size length, and will show them as shorter than 150 bp.**

If the reads overlap a lot, it means you have pseudo-duplicated your data.
You can use tools like VSEARCH, BBMerge, or PEAR to merge the forward and reverse reads that overlap.

| Situation | Problem? |
| --- | --- |
| Positive inner distance (gap) | Normal, no problem |
| Inner distance = 0 | Normal, no problem |
| Negative inner distance, but insert > single read length (overlap only) | Normal, even useful (read merging) |
| Insert < single read length (read runs into the opposite adapter) | **Yes, this is a real problem** — adapter contamination |